

Performance Evaluation of Xception Networks and Short-Time Fourier Transform Spectrograms for Motor Imagery Classification

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Abstract— This study seeks to explore the success of the Xception convolutional neural network and short-time Fourier transform in classifying EEG signals with limited training samples using pre-trained deep transfer learning. So far, most deep learning research has concentrated on one-dimensional time series inputs. Two-dimensional inputs are another excellent way of giving EEG signals to deep learning. This study used a two-dimensional STFT-based method to transform EEG signals into images and Xception for classification. BCI Competition IV dataset 2b, which includes nine participants, was used to evaluate the performance. This study is the first to report the STFT + Xception approach results for classifying motor imagery signals. It achieved good results from a small number of MIEEG data, averaging over 80% for nine subjects with a very low standard deviation of 2.9% between subjects. Applying data augmentation and training from scratch with more data can show more successful results in the future.

Keywords—Motor imagery, Xception network, convolutional neural network, short-time Fourier transform, deep transfer learning

I. INTRODUCTION

Deep neural networks are currently the most popular method for intelligent systems in various applications. They are primarily used in image classification problems. Analysis and classification of biomedical signals is another important research area of increasing importance today. Deep neural networks have great potential to fulfill biomedical signals' automatic feature extraction and classification needs. In this paper, we studied EEG biomedical signals and aimed to detect left/right hand motor imagery (MI) tasks. MI is mainly used in the design of brain-computer interfaces (BCIs), which allow communication between humans and machines. It is essential for partially or fully paralyzed individuals [1]. Recently, it has also started to be used in areas such as drone control.

In the literature, many studies have been conducted to classify MI signals. Besides traditional learning systems, deep learning methods have been increasingly researched in this field. In some of these studies, EEG signals were given as one-dimensional input to deep neural networks. An et al. [2] is one of the first studies on this subject. After de-noising and filtering EEG time series data, they applied fast Fourier transform for time to frequency domain conversion. They classified MI features with boosted single channel deep belief nets. Caglar [3] performed experiments on Graz data set A to classify four MI tasks. They extracted time domain features

using a one-dimensional (1D) convolutional neural network (CNN) and gave them to long short-term memory (LSTM) to obtain high-level representative features. Gouri et al. [4] researched the optimization-enabled deep residual networks and deep learning-based feature fusion. They extracted various features (statistical features, Hjorth's parameters, autoregressive coefficient, etc.) and fused them using mutual information and deep belief networks (DBN).

In other studies, two or more dimensions were used. In many of these studies, EEG signals are expressed in time-frequency space and given as 2D image input to deep neural networks. For example, Kim et al. [5] converted signals to input images using continuous wavelet transform. To handle the BCI illiteracy problem, they proposed a subject-to-subject semantic style transfer network at the feature level. Kaur et al. [6] proposed a time-reassigned multisynchrosqueezing transformation approach and transformed three-channel EEG data to 2D time-frequency representations. After that, they extracted features and classified them employing E-CNNNet hybrid model. Garcia-Moreno et al. [7] sought to develop a low-cost and low-invasive system for identifying left- and right-hand MIs. They constructed a deep learning architecture using LSTM and CNN. They employed a 3D triplet in the input layer, which includes samples, timestamps, and features. Kwak et al. [8] aimed to examine hybrid EEG-fNIRS BCIs. They constructed 3D EEG tensors from EEG signals to handle spatiotemporal information and 3D fNIRS tensors from the fNIRS signals. Then, they employed the fNIRS-guided attention network, a deep learning grounded early fusion structure.

In this study, STFT was first applied to MI-EEG signals of different channels, then MI-related sub-spectrums were extracted and fused. The resulting spectrum images were given as input to deep neural networks. Time, frequency, and channel data of MI-EEG signals were utilized together in constructing inputs. Pre-trained Xception CNN was used for model building to overcome the limitation of insufficient data, a crucial MI-EEG problem. The experimentations were performed on BCI Competition IV - dataset 2b, which includes nine subjects' left/right hand MI-EEG data, and successful results were obtained. The paper is organized as follows. Section 2 presents the dataset with the used 2D image generation technique. Section 3 presents the experimental parameters, literature studies, and comparisons with the previous works. In the last section, conclusions with future recommendations are outlined.

II. MATERIALS AND METHODS

A. Experimental Data

BCI Competition IV dataset 2b: The data set includes nine subjects' left/right hand MI-EEG signals. The cue-based screening paradigm was used. Recordings were taken from C3, Cz, and C4 electrodes with 250 Hz sampling frequency. The signals were filtered between 0.5–100 Hz, and a notch filter was applied at 50 Hz. In data recording, no feedback was provided (screening) in the first two sessions. In contrast, feedback was delivered according to the MI execution performance in the last three sessions. This study was conducted only with data from the first two sessions without feedback (see Fig. 1 for the paradigm). There are 120 EEG samples per session: equal for each MI task. The total dataset includes two sessions and contains 240 samples per subject. The experiments showed a cue expressing the MI task shortly after onset, and the participants were requested to execute the MI for four seconds [9], [10].

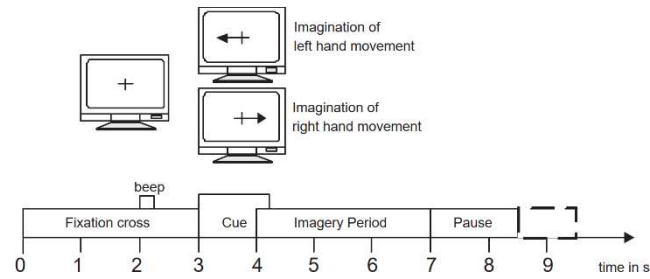


Fig. 1. Timing scheme of the paradigm [9]

B. Methods

2D Image Generation: It is shown by Pfurtscheller et al. [11] that when conducting an MI task, the energy in the mu band decreases. It is named as event-related desynchronization (ERD). Besides, an energy increase in the beta band occurs, and it is named as event-related synchronization (ERS) [12], [13]. Therefore, when left/right hand imagination, ERD and ERS occur in the motor cortex's [12], affecting C4 and C3 electrode locations, respectively. In addition to these channels, the imagery of the hand movements also affects Cz electrode location [12], [13].

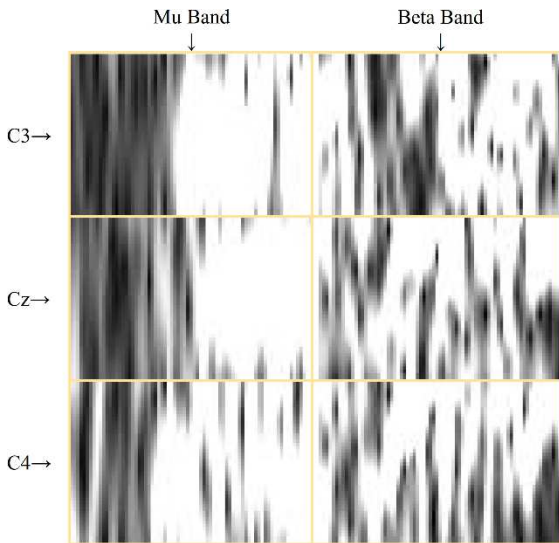


Fig. 2. The image generation process of MI-EEG signals

Before constructing models to classify MI tasks, EEG signals must be processed to remove artifacts, noises, and so on. At the same time, the mu and beta bands should be analyzed to capture ERD and ERS. This study combined spectrograms from different frequency bands and channels to use the effects of different factors, as shown in Fig. 2. In experiments, we used a 3 s length signal (between 4–7 s) corresponding to 750 samples, including the last part of the cue and covering the entire MI period. We calculated the spectrogram with short-time Fourier transform (using Matlab's spectrogram function [14]). In this part, we used the approach proposed by Han et al. [15] with their source code. It was performed using a Hanning with 64 sample window length and 50 sample overlap size. The number of frequency points for computing the discrete Fourier transform was set to 512. This operation produced a 257×50 -sized spectrogram where the numbers 257 and 50 correspond to the frequencies and the times, respectively. Since the information in the mu and beta bands are essential, 16×50 and 29×5 -sized sub-spectrograms of these bands were taken from the spectrogram, respectively. They were then scaled to 50×75 using cubic interpolation. After that, we horizontally combined sub-spectrograms of mu and beta bands, respectively, and constructed 50×150 -sized spectrograms for a one-channel EEG signal. In the last stage, we vertically combined images of C3, Cz, and C4 channels while preserving neighboring information and formed a 150×150 -sized final spectrogram. These spectrograms were then given as input to the Xception CNN. Sample spectrogram images generated from left/right hand MI tasks are illustrated in Fig. 3, respectively.

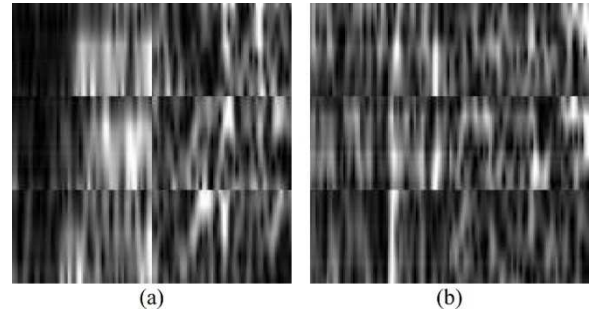


Fig. 3. 150×150 -sized spectrogram images for (a) left and (b) right-hand MI tasks

C. Xception Convolutional Neural Network

Xception network is a deep CNN inspired by Inception and is based on depthwise separable convolution layers. It performs better due to efficient use of model parameters [16]. Combined with transfer learning, it provides fast convergence, high computational efficiency, and good robustness [17]. In addition to these features, this method needs to be researched better in classifying MI-EEG signals and has excellent potential. For these reasons, we used Xception with transfer learning to classify spectrogram images of MI-EEG signals.

Long calibration and training times, subject-to-subject and session-to-session variations are some of the most significant problems in classifying MI-EEG signals. In addition, collecting MI-EEG data is challenging, and most datasets, such as dataset 2b used in this study, need more data [18]. This data scarcity problem limits the use of deep learning methods in MI-EEG studies. The most important candidates for addressing these problems are deep transfer learning and data augmentation. Transfer learning transfers features and

learning models prepared for one specific purpose from one problem domain to another.

In this study, we proposed an MI-EEG classification method that provides high computational efficiency and good robustness by combining Xception with transfer learning. Xception neural networks pre-trained with more than one million images from the ImageNet database were utilized with transfer learning (using MATLAB's neural network toolbox [19]). This network takes inputs of size 299-by-299, and since

it is trained on ImageNet, it can be used to classify 1000 categories of objects. These categories are not related to MIEEG signals and to short-time Fourier transform spectrogram. For this reason, we fine-tuned the final layers and employed transfer learning to classify MI-EEG spectrogram images. We constructed a new fully connected layer to replace the previous one and updated the classification layer to include two output nodes.

TABLE I. COMPARISON WITH STUDIES ON DATASET 2B TO CLASSIFY LEFT/RIGHT HAND MI TASKS (IN %)

Subjects	FBCSP with "whole" segment [20]	FBCSP with "2s1o" segment [20]	IS-CBAM-CNN [21]	DivFE without transformation [22]	DivFE with transformation [22]	STFT with Xception Network
B01	75.5	80	80.3±1.5	95	95.3	81.67±6.24
B02	63.64	60.61	75.0±1.8	79.5	78.5	80.00±4.49
B03	51.12	55.9	67.7±2.6	82.9	73.7	77.92±4.58
B04	80.49	82.11	95.4±0.6	98	92.3	86.92±5.22
B05	70.71	76.36	88.3±1.5	91.4	96	78.85±4.63
B06	75.78	79.81	80.0±1.7	93.5	88.8	78.75±6.31
B07	64.22	66.97	73.7±2.2	90.6	82.9	77.50±6.24
B08	58.56	65.32	77.4±2.0	86.3	78.7	82.14±5.30
B09	67.03	69.51	78.6±2.1	80.5	80	77.08±3.84
Average	67.45	70.73	79.6	88.6	85.1	80.09
Std. dev.	8.73	8.80	7.67	6.25	7.7	2.93

III. EXPERIMENTS AND RESULTS

In this paper, we only used the first two sessions' data, without feedback. Because these sessions contain more challenging MI-EEG data. The data from these two sessions were, first merged. All EEG trials were converted into images for, each subject and evaluated with a 10-fold CV. Classification, performances were obtained for accuracy and kappa. In this, paper, we used Xception pre-trained CNNs to classify a new, collection of spectrogram images. It was pre-trained on the, ImageNet. To avoid overfitting, an early stopping strategy was, used in the training. In fine-tuning, the following parameters, were used. 1e-3, 1e-4, and 1e-5 were used as initial learning, rates. Adam and stochastic gradient descent with momentum, optimizers were used to minimize the error. The maximum, number of epochs was set at 50. We enabled the training for, mini-batch sizes of 2 and 4. Different parameter combinations, in diverse folds and subjects yielded the most successful, results. Therefore, the best performance values were used, when presenting the results (see Sec. II-C for details).

Comparisons with the literature were made with the studies, [20], [21], and [22], which selected MI-EEG data without, feedback (first two sessions) as the dataset. These are recent, works that aim to solve various issues related to MI-EEG, classification problems. The comparative results obtained for, the classification of two-class left/right hand MI tasks are, shown in Table I. The approaches proposed in the literature, are as follows in detail. Nguyen et al. [20] developed a novel, method to discover an optimal combination of time segments, and feature extractors using short-window segments. Features, were extracted using CSP and its variations and classified, using Linear Discriminant Analysis. They discovered a negative correlation between the

performance and subject-specific frequency bands. They found that the model's accuracy increases with narrower and more focused frequency ranges. Chen et al. [21] et al. mentioned three problems to enhance the classification. These are the non-stationary nature, excitation occurrence's temporal localization, and frequency band distribution characteristics. They used wavelet transform to convert signals to images containing energy values as well as time time-frequency domain information. After that, they developed a new method using time-frequency image subtraction (IS) to synthesize the input. A Convolutional Block Attention Module (CBAM) was then used to extract spatial and channel information with these inputs. The proposed IS-CBAM- CNN framework achieved 79.6±1.8% average accuracy and 0.592±0.036 kappa values across subjects. [22] have focused on data augmentation to succeed in problems with small data sets such as MI-EEG and proposed the Divergence-based Feature Extractor (DivFE) network. They tried to increase the success of DNNs with fewer nodes and hyperparameters. Following the last layer of the CNN, a minimum distance network (MDN) was employed for classification, which takes the proposed feature extractor's output as input. In this paper, the MI EEG epoch matrices of the channels were employed as the input of the feature extractor, not the image-based inputs.

The proposed approach obtained 80.09±2.93% average accuracy across all subjects. Regarding subjects, the highest success rate was 86.92% for B04, and the lowest was 77.08% for B09. The average standard deviation between folds was 5.1% for all subjects. Nguyen et al. [20] achieved 70.73±8.80% average accuracy across subjects using Filter Bank CSP (FBCSP) with 2-s length and 1-s overlapping ("2s1o") time segments. They also obtained 67.45±8.73% accuracy when using the whole MI segments. The results for

“2s1o” were slightly higher than the “whole” segment, except for one subject. The results showed that FBCSP successfully extracts features, and “2s1o” segments could be more appropriate for online BCIs. Chen et al. [21] achieved $79.6 \pm 1.8\%$ average accuracy and 0.592 ± 0.036 kappa using the IS-CBAM-CNN framework. The third subject showed lower results than the others, with a $67.7 \pm 2.6\%$ accuracy. Besides, performance decline was observed when either IS or CBAM was removed or replaced. When using data augmentation with no transformation stage, the DivFE [22] showed an average of $88.6 \pm 6.25\%$ with 0.772 kappa. When the transformation stage was used, it showed an average accuracy of $85.1 \pm 7.7\%$ and a kappa of 0.702. Data augmentation has contributed significantly to these results. Without augmentation, the proposed method with transformation and without transformation stage showed low average results of 70.6% and 68.5%, respectively.

The proposed method outperformed FBCSP [20] and IS-CBAM-CNN [21], with improvements in the 0.49–12.45% range. It is more successful in FBCSP with “whole” segment [20] for all subjects, FBCSP with “2s1o” segment [20] for all subjects except B06, IS-CBAM-CNN [21] for all subjects except B04–B06 and B09. However, on average, 6.7% lower results were obtained compared to DivFE-based methods [22]. The method’s success is due to data augmentation, which increased the classification accuracy by about 14.5%. In our study, we applied only transfer learning without data augmentation and built the model with a small number of data. However, in addition to increasing the average success above 80%, we achieved remarkable results, especially in the standard deviation of 2.93% between nine subjects, which is very good compared to the literature. The results can be further improved by applying augmentation as in the [22] or deep learning from scratch. Besides, more successful results can be obtained if the entire MI signal is segmented into “2s1o” as in [20] with a length of 2 s and an overlap of 1 s.

IV. CONCLUSION

In this study, we tried to classify different MI tasks with a pre-trained Xception CNN with little data. Image inputs generated by STFT using different electrode positions and frequency bands were given as input to this network. This study has not considered a large number of training data, and no training was done from scratch. Nevertheless, the average results are quite good, and the standard deviations across subjects are relatively low compared to the literature. The classification performance can be further improved by data augmentation and training from scratch with more data. Besides, further investigations are needed to research the processing of MI-EEG signals in deep neural network aspects in more detail. In particular, extracting excellent features from complex MI signals is very difficult. These difficulties increase as subject-to-subject and session-to-session variations were also involved. The present study obtained the best results in different subjects and folds with different hyperparameters. Therefore, deep learning problems such as tuning hyper-parameters should also be investigated. In addition, studies should be carried out for systems that can operate in real-world scenarios, which should be the goal.

ACKNOWLEDGEMENT

The authors would like to thank the Knowledge Discovery Institute, Graz University of Technology, for

sharing the dataset. We also would like to thank the Council of Higher Education of Turkey.

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